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Batch: 2025-29

ASSIGNMENT No: 12

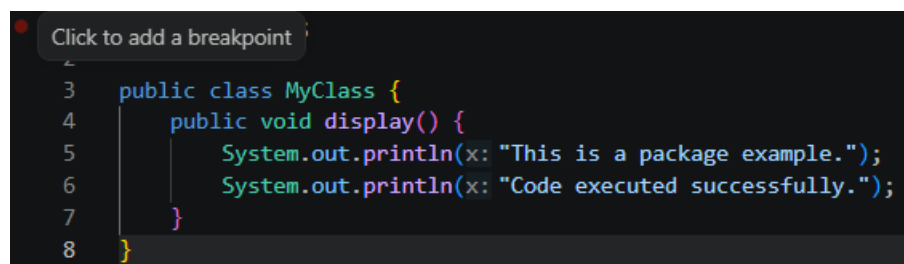
TITLE : Write a Java program to create and use user-defined packages.

Problem Statement: Write a Java program to create and use user-defined packages.

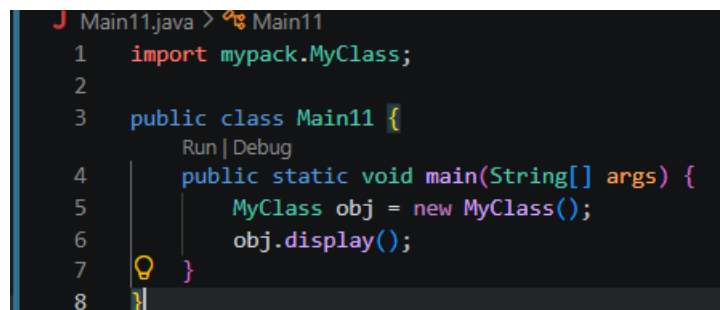
Learning Objectives: To create and use user-defined packages for organizing Java programs.

Theory: Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

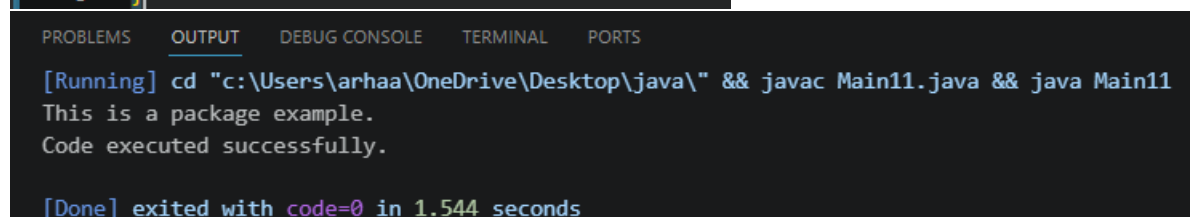
Code:



```
Click to add a breakpoint
3 public class MyClass {
4     public void display() {
5         System.out.println(x: "This is a package example.");
6         System.out.println(x: "Code executed successfully.");
7     }
8 }
```



```
Main11.java > Main11
1 import mypack.MyClass;
2
3 public class Main11 {
4     Run | Debug
5     public static void main(String[] args) {
6         MyClass obj = new MyClass();
7         obj.display();
8     }
9 }
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac Main11.java && java Main11
This is a package example.
Code executed successfully.
[Done] exited with code=0 in 1.544 seconds
```

Conclusion

Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```

1 package details;
2
3 public class StdTls {
4     String name;
5     int age;
6     String branch;
7
8     public StdTls(String name, int age, String branch) {
9         this.name = name;
10        this.age = age;
11        this.branch = branch;
12    }
13
14    public void display() {
15        System.out.println("Name: " + name);
16        System.out.println("Age: " + age);
17        System.out.println("Branch: " + branch);
18    }
19 }

```

```

1 package details;
2
3 public class Tdtls {
4     String name;
5     int tid;
6     String dept;
7
8     public Tdtls(String name, int tid, String dept) {
9         this.name = name;
10        this.tid = tid;
11        this.dept = dept;
12    }
13
14    public void display() {
15        System.out.println("Name: " + name);
16        System.out.println("Teacher ID: " + tid);
17        System.out.println("Department: " + dept);
18    }
19 }

```

```

1 import details.StdTls;
2 import details.Tdtls;
3
4 public class Main12 {
5     Run | Debug
6     public static void main(String args[]) {
7         StdTls student1 = new StdTls(name: "Arhaan", age: 18, branch: "CSE");
8         StdTls student2 = new StdTls(name: "Rachit", age: 19, branch: "ENTC");
9
10        Tdtls t1 = new Tdtls(name: "Dr Ketan Kotecha", tid: 10, dept: "AI and ML");
11        Tdtls t2 = new Tdtls(name: "Dr Rupali Nagar", tid: 55, dept: "Physics");
12
13        System.out.println(x: "Student 1 Details:");
14        student1.display();
15        System.out.println(x: "\nStudent 2 Details:");
16        student2.display();
17
18        System.out.println(x: "\nTeacher 1 Details:");
19        t1.display();
20        System.out.println(x: "\nTeacher 2 Details:");
21        t2.display();
22    }

```

```

[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac Main12.java && java Main12
Student 1 Details:
Name: Arhaan
Age: 18
Branch: CSE

Student 2 Details:
Name: Rachit
Age: 19
Branch: ENTC

Teacher 1 Details:
Name: Dr Ketan Kotecha
Teacher ID: 10
Department: AI and ML

Teacher 2 Details:
Name: Dr Rupali Nagar
Teacher ID: 55
Department: Physics

[Done] exited with code=0 in 1.434 seconds

```

2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
1 package details;
2
3 public class Tdtls {
4     String name;
5     int tid;
6     String dept;
7
8     public Tdtls(String name, int tid, String dept) {
9         this.name = name;
10        this.tid = tid;
11        this.dept = dept;
12    }
13
14    public void display() {
15        System.out.println("Name: " + name);
16        System.out.println("Teacher ID: " + tid);
17        System.out.println("Department: " + dept);
18    }
19 }
```

```
J Main13.java > ...
1 import library.Bdtls;
2
3 public class Main13 {
4     Run | Debug
5     public static void main(String[] args) {
6         Bdtls book1 = new Bdtls(Bid: 101, author: "Tolkien", price: 600);
7         Bdtls book2 = new Bdtls(Bid: 150, author: "GRR Martin", price: 400);
8         Bdtls book3 = new Bdtls(Bid: 110, author: "Homer", price: 1000);
9
10        System.out.println(x: "Book 1 Details:");
11        book1.display();
12
13        System.out.println(x: "\nBook 2 Details:");
14        book2.display();
15
16        System.out.println(x: "\nBook 3 Details:");
17        book3.display();
18    }
19 }
```

```
[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac Main13.java && java Main13
Book 1 Details:
Book ID: 101
Author: Tolkien
Price: 600

Book 2 Details:
Book ID: 150
Author: GRR Martin
Price: 400

Book 3 Details:
Book ID: 110
Author: Homer
Price: 1000

[Done] exited with code=0 in 1.174 seconds
```